Journal of Food Composition and Analysis 109 (2022) 104499

Contents lists available at ScienceDirect

Journal of Food Composition and Analysis

journal homepage: www.elsevier.com/locate/jfca

Review article

Uncover the flavor code of strong-aroma baijiu: Research progress on the revelation of aroma compounds in strong-aroma baijiu by means of modern separation technology and molecular sensory evaluation

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ARTICLE INFO

Keywords:
Strong-aroma baijiu
Trace components
Flavor
Sensory evaluation
Qualitative analysis
Quantitative analysis

ABSTRACT

Baijiu is widely regarded as the national liquor of China and occupies a vital position in China's food industry. In 2020, the total sales revenue of baijiu industry reached 583.639 billion yuan. Strong-aroma baijiu is one of the four basic aroma baijiu types, and is popular among consumers because of its pleasant flavor. Thus, the researches on the trace components in strong-aroma baijiu and their influences on the aroma profile of strong-aroma baijiu have been carried out gradually. This study mainly introduced the technological process of strong-aroma baijiu, summarized recent progress in the research on the trace components in strong-aroma baijiu, especially for the influences of aroma compounds on the aroma profile of strong-aroma baijiu. This study aims to provide ideas for the research direction of strong-aroma baijiu, and lay a foundation for the improvement of related standards and the high-quality development of the baijiu industry in the future.

1. Introduction

China has a long history of alcoholic beverages brewing for more than 9000 years. Of note, baijiu is the most popular alcoholic beverages in China, and has a unique position in the traditional Chinese culture. It has almost penetrated into every aspect of social life, and become a medium for human beings to express their feelings and communicate with each other. As the saying goes, "No banquet without alcoholic beverage." Since ancient times, there has been no lack of alcoholic beverages at all kinds of banquets in China.

According to the literature, the earliest alcoholic beverage of China was made from rice, hawthorn, grapes, honey and so on (McGovern et al., 2004). Based on, rice wine, huangjiu and baijiu are unique alcoholic beverage of China. Around 9000 years ago, rice wine appeared. Subsequently, after 2000 years of technological succession, huangjiu began to appear on the stage of history. Due to more than 7000 years of brewing and drinking history, huangjiu is collectively referred to as the world's three oldest liquors together with wine and beer. About 5000 years later, the appearance of distillation technology promoted the emergence and prosperity of baijiu. As reported, baijiu had a history of more than 2000 years and was one of the earliest distilled liquors all over the world. In general, baijiu is colorless, transparent, fragrant and full-bodied with a mellow taste. Because of its long history, high alcohol concentration and peculiar flavor, baijiu is collectively known as the six world-renowned distilled liquors with whiskey, brandy, vodka, gin and rum.

Among the six most famous distilled liquors, baijiu is the one with the longest history and fermentation cycle, the most complex technology, the largest output and the fewest producing countries (only in China). Compared with other distilled liquors, baijiu is the only one that uses Jiuqu (natural inoculation, multi-strains) as saccharification starter, while other distilled liquors are mainly fermented by yeast (single strains) (Jin et al., 2017). Moreover, the raw materials of baijiu are also more abundant, including sorghum, corn, rice, wheat, glutinous rice and other starchy grain cereals. While other kinds of distilled liquors usually use single raw materials for fermentation. For example, brandy uses grapes or other fruits as raw materials, and rum uses sugarcane as raw materials. In addition, the process of baijiu production is open, and microorganisms naturally colonize on solid baijiu raw materials to participate in fermentation process. That is, baijiu production is a solid spontaneous fermentation (Wang et al., 2020b). Besides, the production of baijiu is different from that of other distilled liquors because it combines the two distinctive processes of saccharification and fermentation.

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https://doi.org/10.1016/j.jfca.2022.104499
Received 2 December 2021; Received in revised form 15 February 2022; Accepted 28 February 2022
Available online 4 March 2022
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Nomenclature

DI	direct injection
LLE	liquid–liquid extraction
LLME	liquid-liquid microextraction
SPE	solid-phase extraction
SPME	solid-phase micro-extraction
HS-SPME	headspace solid-phase micro-extraction
SBSE	stir bar sorptive extraction
SDE	simultaneous distillation extraction
MAE	microwave-assisted extraction
PT	purge and trap
MS	mass-spectrometry
FID	flame ionization detecto
FPD	flame photometric detector
PFPD	pulsed flame photometric detector
NPD	nitrogen phosphorus detector
SCD	sulfur chemiluminescence detector
GC×GC	comprehensive two-dimensional gas chromatography
GC×GC-TOFMS	comprehensive two-dimensional gas chromatography-time-of-flight mass spectrometry
HPLC	high-performance liquid chromatography
GC–FTD	gas-flame thermal ion detector
SIM	selected ion monitor
GC-O	gas chromatography-olfactometry
OAV	odor activity value
AEDA	aroma extract dilution analysis
FD	flavor dilution

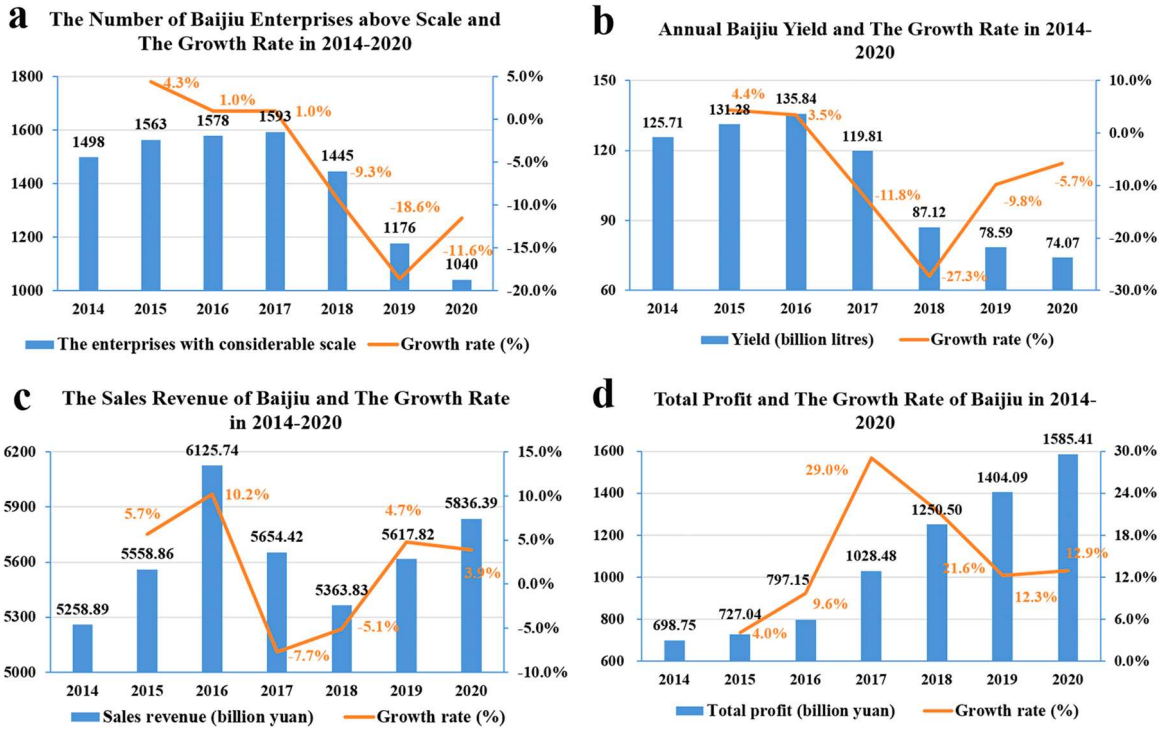

Fig. 1. Overview of baijiu industry from 2014 to 2020. The number of baijiu enterprises above scale and the corresponding growth rate (%) from 2014 to 2020 (a). Annual baijiu yield (billion litres) and the corresponding growth rate (%) from 2014 to 2020 (b). The sales revenue of baijiu (billion yuan) and the corresponding growth rate (%) from 2014 to 2020 (c). Total profit (billion yuan) of baijiu and the corresponding growth rate (%) from 2014 to 2020 (d).

During the fermentation process of baijiu, conversions of starch to sugar and sugar to alcohol take place simultaneously (Liu and Sun, 2018). In the production of other distilled liquors, such as whisky and vodka, the above two reactions occur stepwise, or conversion of sugar to alcohol takes place alone, such as in rum production.

Nowadays, baijiu occupies an important position in the Chinese food industry. According to the data of National Bureau of Statistics (as shown in Fig. 1), in 2020, the number of baijiu enterprises above scale and the annual yield were 1040 and 7.407 billion liters, respectively. Meanwhile, the total sales revenue of baijiu industry reached 583.639 billion yuan, and the total annual profit was 158.541 billion yuan. In general, the resources of baijiu industry are gradually integrated, the concentration of baijiu industry is increasing, and the production capacity is gradually down-regulated. Whilst the sales revenue has fluctuation, the annual total profit has gradually increased in recent years, which reveals that the added value of baijiu products is gradually improving. Besides, the reduction of production capacity does not have much impact on the total profit, indicating that the output of mid-to-high end baijiu is increasing. In the future, high-quality baijiu will become the development direction of baijiu industry.

As a kind of daily alcohol beverage, the flavor is the key factor for consumers to choose baijiu. Due to the complex raw materials, unique technology and co-fermentation of multiple strains, the components in baijiu are extremely complex. From the perspective of chemical composition, there are 97–98% water and ethanol in baijiu, and only 2–3% are trace components. Although the contents of these trace components are very low, they have a significant contribution to the aroma characteristics of baijiu (Fan et al., 2014; Li et al., 2019). According to the statistical results of our research group, there are more than 2020 kinds of trace components in baijiu have been detected (as shown in Fig. 2), including 510 esters, 249 alcohols, 140 acids, 18 lactones, 102 aldehydes, 160 ketones, 48 acetals, 82 sulfur compounds, 155 nitrogen

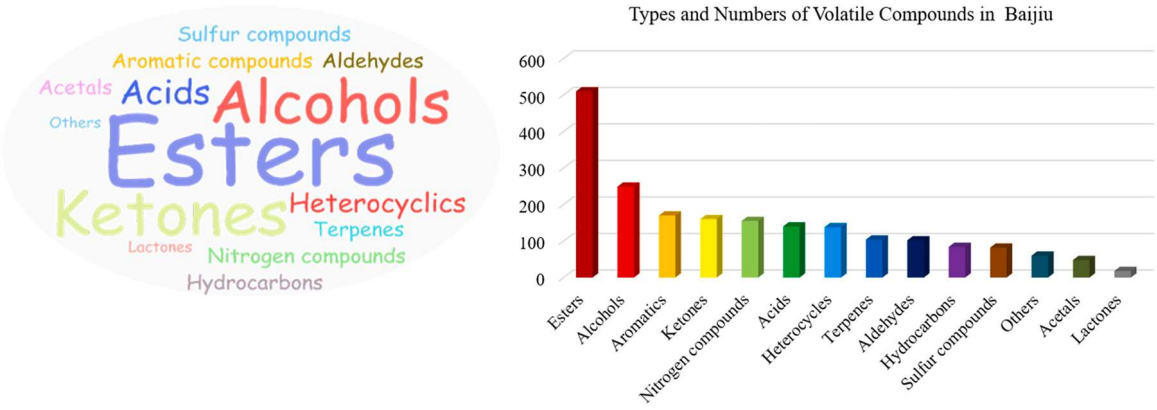

Fig. 2. Types and numbers of trace components in baijiu. There are 2020 kinds of trace components in baijiu have been detected. The wordle map of trace components on the left indicates the proportions of different types of trace components and the bar chart on the right shows the specific number of different types of compounds in baijiu.

compounds, 138 heterocyclic compounds, 170 aromatic compounds, 84 hydrocarbons, 104 terpenes, and 60 others. Among them, the number of ester compounds is the largest, and most of them have pleasant aromas such as fruity and sweet aromas, which have an important impact on the flavor of baijiu.

The condition of baijiu brewing in distinct regions are quite different. Therefore, the trace components of baijiu in different regions have diversity, which leads to the various aroma characteristic of baijiu. According to its main aroma characteristics, baijiu can be artificially classified into twelve aroma types, that is strong-aroma, sauce-aroma (jiang-aroma), light-aroma, rice-aroma, mixed-aroma, sesame-aroma, fen-aroma, herbal-aroma (dong-aroma), fuyu-aroma, te-aroma, chi-aroma, and laobaigan-aroma (Zheng and Han, 2016). Among them, strong, light, sauce, and rice aroma are widely recognized as the four basic aroma types of baijiu. The other eight aroma types are derived from the four basic aroma types. As one of the four basic aroma types, strong-aroma baijiu is deeply loved by the majority of drinkers because of its harmonious flavor. And its sales share accounts for about 70% of the total yield or market share of baijiu in China, ranking first in the list of various flavors of baijiu for many years.

The trace components in baijiu have great influence on baijiu flavor. Therefore, the researches on the trace components in baijiu, especially for strong-aroma baijiu, has been carried out and some progress has been made. Based on this, this review systematically combs the extraction methods and detection technologies of trace components in baijiu, and summarizes the research progress of the detected trace components and aroma compounds in strong-aroma baijiu. It is hoped that this paper will lay a theoretical basis for the change rule, metabolic pathway and generation mechanism of the trace components of strong-aroma baijiu during the fermentation process, and provide ideas for the transformation and upgrading of the baijiu industry.

2. Introduction of strong-aroma baijiu

Strong-aroma baijiu is one of the four basic aroma types, and the typical characteristics of strong-aroma baijiu are strong cellar aroma, pure and harmonious, coordinated aroma and long aftertaste. Because of its harmonious flavor, it is deeply loved by the majority of drinkers. And

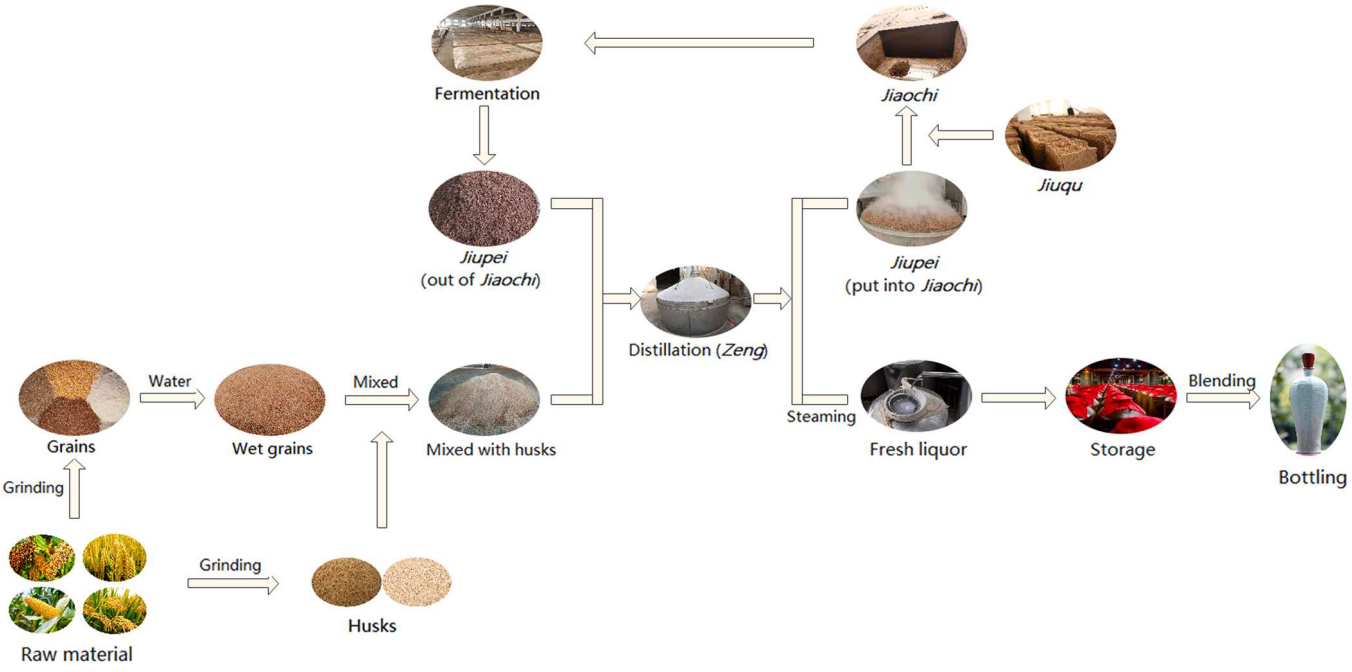

Fig. 3. Traditional processing of strong-aroma baijiu production. Sorghum, rice, glutinous rice, wheat and corn are usually used as raw materials for baijiu, and a unique solid-state fermentation and distillation brewing process is adopted.

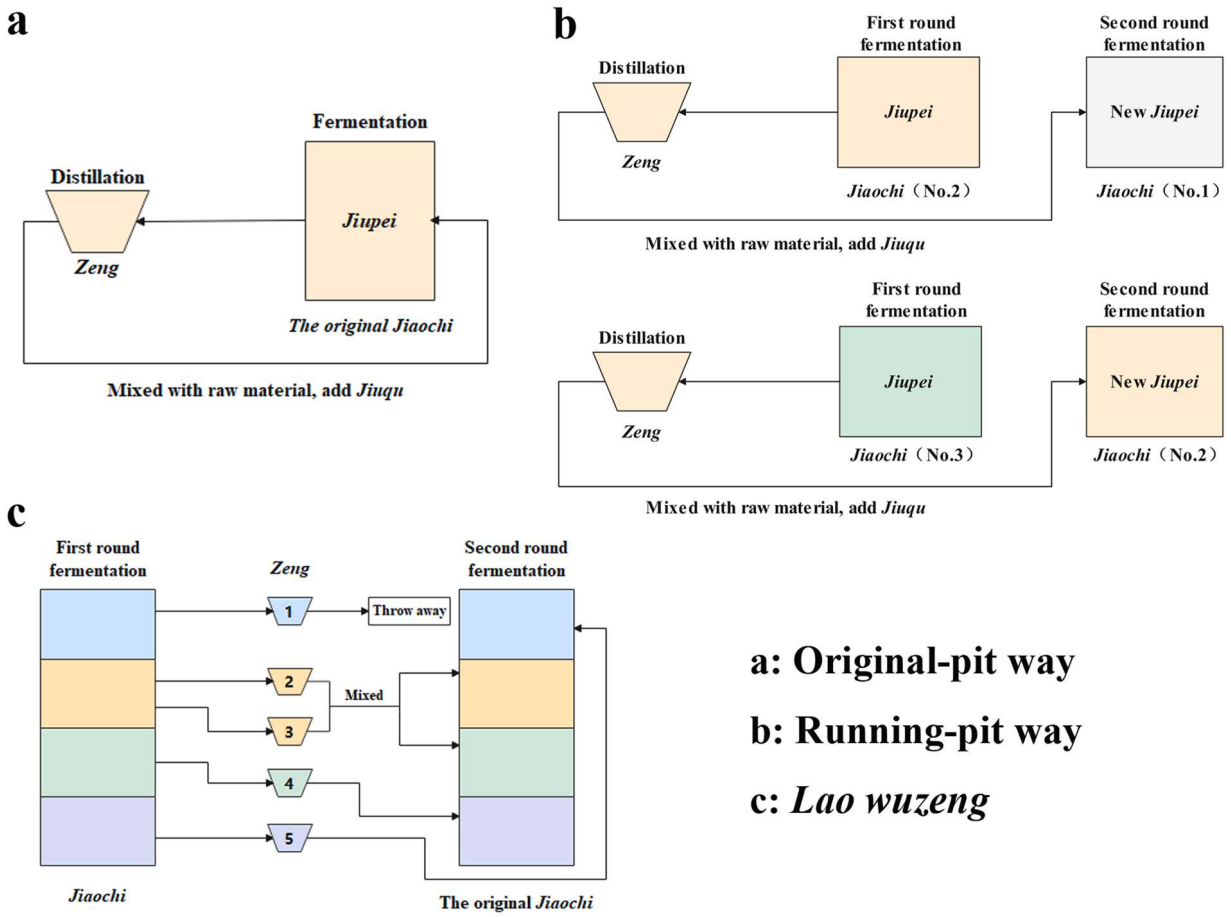

Fig. 4. The different fermentation and distillation technology for strong-aroma baijiu. Two categories of fermentation technology are applied in the brewing of strong-aroma baijiu, that is Original-pit way (a) and Running-pit way (b). And the commonly used technology of distillation in baijiu production is *Lao wuzeng* (c).

its sales share accounts for about 70% of the total yield or market share of baijiu in China, ranking first in the list of various flavors of baijiu for many years. Recently, strong-aroma baijiu become a hot topic of study. Strong-aroma baijiu was originally called Luzhou-flavor baijiu, named after the characteristics of Luzhou Laojiao (a famous brand of strong-aroma baijiu), and was called strong-aroma baijiu since 1980 s.

The definition of strong-aroma baijiu according to the national standard is that strong-aroma baijiu is made of grains as raw material, using strong-aroma-*Daqu* as saccharification starter. And then it will be finished through solid fermentation in *Jiaochi* (a saccharification and fermentation vessel), solid distillation, aging and blending. The traditional processing of strong-aroma baijiu production is shown in Fig. 3. The raw materials should first be crushed and pretreated, and mixed uniformly. Afterwards, the mixed grains are put into a special device for gelatinization with water steam, which is called the *Zeng*. Then, the steamed grains are cooled and added with *Jiuqu*, subsequently, they are put into the *Jiaochi* for fermentation. The solid-state fermentation period of strong-aroma type of baijiu is generally 30 ~ 90 days (Hong et al., 2021b). After that, *Jiupei* (fermented grains) is mixed with rice husk, distilled under solid-state conditions to produce fresh baijiu. The fresh baijiu is then stored and aged in a pottery or stainless-steel vessel. After aging for several years, the stored baijiu is blended to obtain differently graded products.

Most strong-aroma baijiu use sorghum, mixed with rice, glutinous rice, wheat and corn, as raw materials. According to the use of raw materials, strong-aroma baijiu can be divided into single-grain strong-aroma baijiu and multiple-grains strong-aroma baijiu (Li, 2009). Single-grain strong-aroma baijiu is fermented with sorghum as the single raw material, with Luzhou Laojiao as the typical brand. Multiple-grains strong-aroma baijiu is typically represented by Wuliangye, which is fermented from five grains of wheat, rice, corn, sorghum, and glutinous rice. Multi-grain baijiu gives full play to the advantages of various raw materials. The interaction of various grains provides a comprehensive material basis for enriching the flavor of baijiu. In comparison, single-grain strong-aroma baijiu has a purer taste. Brewing materials of baijiu generally require a high carbohydrate content, appropriate protein and tannin contents to meet the needs of microorganisms. In addition, the raw materials should be easy to store, and the water content should be strictly controlled to prevent mildew and deterioration.

On the basis of the difference of *Jiuqu*, strong-aroma baijiu can be divided into *Daqu* strong-aroma baijiu and *Fuqu* strong-aroma baijiu, among which *Daqu* strong-aroma baijiu with the highest yield is the most famous. *Daqu* also uses grains (such as wheat, barley, and peas) as raw materials. During the production process of *Daqu*, wetting wheat and other raw materials with water are pulverized to fine particles by a mixer. Then, the crushed raw material is mixed with water and then put into a crimping machine to press into a *Jiuqu* block. Subsequently, the prepared *Jiuqu* block is put into the fermentation room to be cultured. In the process of culture, the *Jiuqu* block is turned over several times in order to make microorganisms colonize uniformly on *Jiuqu* block. Finally, cultured *Jiuqu* block is moved out of fermentation room and stored in the finished storehouse which is ventilated and dried. It is worth noting that the *Jiuqu* block is usually crushed into powder when used.

Normally, two categories of fermentation technology are applied in the brewing of strong-aroma baijiu, that is Running-pit way and Original-pit way (Lai, 2005). These two fermentation processes are mainly produced in Sichuan province. And the commonly used

Fig. 5. The distribution of strong-aroma baijiu in China. Strong-aroma baijiu is mainly distributed in Sichuan region, Jiang-huai region and Northern region, and the main brands are Luzhou Laojiao, Wuliangye, Gujinggong Jiu, Pancheng Boiler and so on.

technology of distillation in baijiu production is *Lao wuzeng*. The simple diagrams of the three different technology are respectively shown in Fig. 4.

Original-pit way means that *Jiupei*, after being layered distilled and adding raw materials and *Jiuqu*, is still put back into the original *Jiaochi* for the next round fermentation. Significantly, *Jiupei* taken out from *Jiaochi* should be stacked in layers. Running-pit way refers to preparing an empty *Jiaochi* (No. 1) at first. Then *Jiupei* in *Jiaochi* (No. 2) is taken out for distillation, and loaded into the prepared *Jiaochi* (No. 1) for the next round fermentation. Hence, *Jiaochi* (No. 2) is empty. Afterwards, *Jiupei* in *Jiaochi* (No. 3) is taken out for distillation, and put into *Jiaochi* (No. 2), thus, *Jiaochi* (No. 3) becomes empty. Such a cycle is named as Running-pit way. As a traditional distillation technology, *Lao wuzeng* means that the *Jiupei* (from the *Jiaochi*) mixed with raw materials are distilled, steamed and gelatinized at the same time in *Zeng*. The *Jiupei* in *Jiaochi* are divided into four layers, namely covering layer, upper layer, middle layer, and base layer. After a round of fermentation, the covering layer is distilled and thrown away, and the upper layer is mixed with new raw materials and husks. Then the mixture is divided into new upper layer and middle layer in the next round of fermentation. The middle layer becomes the new base layer, and the base layer will become new covering layer, which is abandoned after fermentation. This cyclical distillation technology is called *Lao wuzeng*.

Of note, the main difference between the two kinds of fermentation technology is whether to use the original *Jiaochi* for the next round of fermentation. Original-pit way always uses the original *Jiaochi* for the next round of fermentation, while the running-pit way uses a new one for the next round of fermentation. Therefore, the original-pit way emphasizes maintaining the grade quality of the *Jiaochi* and pit mud, thus, stabilizes the unity of the brewing flavor. In contrast, the running-pit way involves multiple *Jiaochi* in the fermentation process, and the flavor characteristic of different *Jiaochi* will complement each other, so the balance and consistency of baijiu flavor is better.

Strong-aroma baijiu can be divided into three types according to flavor characteristics (Hong et al., 2021a). They are Sichuan region strong-aroma baijiu, Jiang-huai region strong-aroma baijiu, and Northern region strong-aroma baijiu. Jiang-huai region strong-aroma baijiu is commonly known in areas such as Jiangsu province, Anhui province and so on. Northern region mainly from Inner Mongolia and Hebei province and so forth. Jiang-huai region is represented by Yanghe Daqu and Gujinggong Jiu. The main representative brands of Sichuan region are Luzhou Laojiao, Wuliangye, Jiannanchun and so on. Northern region includes Pancheng Boiler, Hetaowang Jiu, and Longquanchun Jiu (see Fig. 5). Jianghuai region and Sichuan region belong to the subtropical monsoon climate zone, and are adjacent to the Yangtze and Yellow Rivers. They have a relatively humid climate, abundant water sources and a wide variety of microorganisms. The Northern region is located in the north of the temperate monsoon with relatively weak diversity of microorganisms.

Due to the different impacts of geographical environment, climate, humans and animals on the ecological environment in various regions, it brings the different microflora and many other factors accordingly (Fan et al., 2021; Xu et al., 2022). In addition, factors such as raw materials, saccharification and starter, brewing technology, fermentation and storage equipment, blending technology level are not the same, thus, there are subtle or obvious differences in the styles of strong-aroma baijiu produced in various regions. The flavor of strong-aroma baijiu in Jianghuai and Sichuan region is relatively rich and powerful, but the taste of Northern strong-aroma baijiu is relatively mellow and clear.

The composition of strong-aroma baijiu is very abundant because of the diverse raw materials and complicated technology. At present, 861 kinds of trace components have been found in strong-aroma baijiu (as

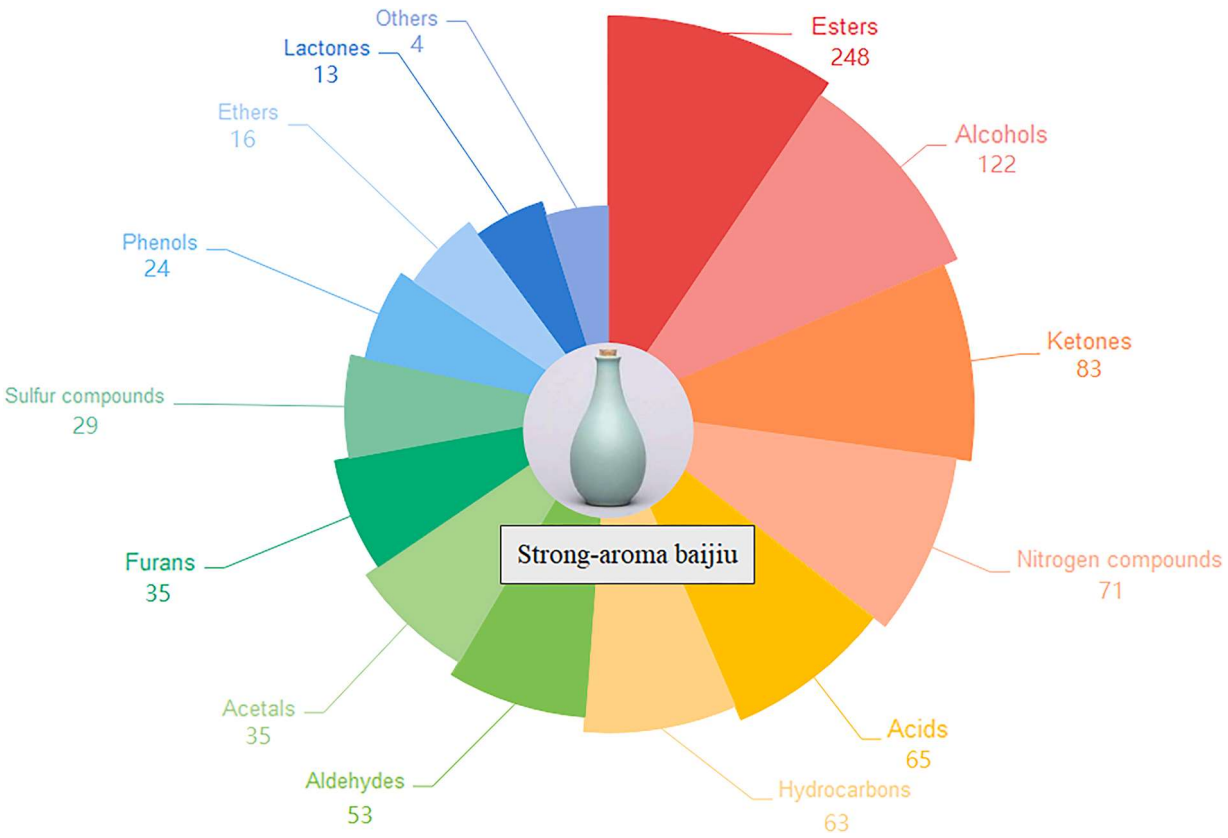

Fig. 6. Types and numbers of trace components in strong-aroma baijiu. At present, 861 kinds of trace components have been found in strong-aroma baijiu, including 248 esters, 122 alcohols, 65 acids and 53 aldehydes, 83 ketones, 35 acetals, 29 sulfur compounds, 71 nitrogen compounds, 24 phenols, 16 ethers, 63 hydrocarbons, 35 furans, 13 lactones, and 4 others.

shown in Fig. 6), including 248 esters, 122 alcohols, 65 acids and 53 aldehydes, 83 ketones, 35 acetals, 29 sulfur compounds, 71 nitrogen compounds, 24 phenols, 16 ethers, 63 hydrocarbons, 35 furans, 13 lactones, and 4 others (Sun, 2019). Esters have the largest content among the various components in strong-aroma baijiu. Among them, ethyl hexanoate, with high content and low aroma threshold (0.76 mg/L), has the aroma characteristics of sweet and fruit, which makes an important contribution to the main aroma characteristics of strong-aroma baijiu. In addition, ethyl lactate, ethyl acetate, and ethyl butanoate, which have high content as well, are collectively known as the "four major esters" with ethyl hexanoate of strong-aroma baijiu. Moreover, the content of organic acids is next to ester, with a total content of 140 mg/100 mL. Among them, acetic acid, hexanoic acid, butyric acid and lactic acid are the four main kinds of acids in baijiu. The duration of taste of baijiu is closely related to some organic acids with higher boiling points. Alcohol is another kind of important aroma compound in strong-aroma baijiu, with a total content of about 103 mg/100 mL. Alcohols generally contribute fruity and sweet aroma to the flavor of baijiu. However, high content of alcohols will make the aroma and taste worse. For instance, high content of 2-butanol, 2-methyl-1-propanol, and 1-butanol will make baijiu have obvious bitter taste.

3. Extraction method of trace components in baijiu

The trace components in baijiu contribute to the flavor quality of baijiu. Hence, the research on the trace components in baijiu is gradually carried out, and the extraction of the trace components is the first step. The trace components in baijiu are very plentiful and the contents of the compounds are varied, and there are different pretreatment methods being used to extract the different target compounds. Nowadays, a variety of extraction methods have been used in baijiu flavor analysis, such

Table 1
Extraction methods of trace components in baijiu.

Extraction method	Advantages	Disadvantages	Applicable components
Direct injection (DI)	Fast, simple, avoiding the loss of trace components during sample preparation	Damage to the chromatographic column and detector, high LOD (limit of detection), few kinds of components detected	High concentration skeleton components
Liquid-liquid extraction (LLE)	Great capability for enrichment, high recovery rate and wide application range	Complicated operation, large volume of organic solvents and large sample loss	Most of the trace components
Liquid-liquid microextraction (LLME)	Less sample consumption, shorter pretreatment time, and relatively simple steps	Unsuitable for the extraction of some low-content components	High-content and high-polarity components
Solid-phase extraction (SPE)	Low consumption of sample and solvent, good universality and selectivity	Relatively weak enrichment capacity, high cost	Most of the trace components
Solid phase microextraction (SPME)	Solvent-free, simple operation, highly automated	Limited enrichment capacity, poor repeatability and expensive extraction fiber	Volatile or semi-volatile components
Stir bar sorptive extraction (SBSE)	Larger adsorption capacity than SPME, high sensitivity and selectivity	Coating extraction material is limited and there are certain limitations	Most of the trace components

as direct injection, liquid-liquid extraction, solid-phase extraction, stir bar sorptive extraction, etc. And each method has its own characteristics (as shown in Table 1).

Direct injection (DI) is the briefest method, and the sample is not processed or just simple processed. Specifically, DI is fast and simple attributed to the fact that DI omits the tedious steps of sample pretreatment and avoids the loss of trace components to a certain extent during sample preparation. In 1997, Cai et al. detected the trace components of Moutai Jiu by direct injection, and a total of 73 trace components were found (Cai et al., 1997). In 2018, direct injection combined with gas chromatography–mass spectrometry (GC–MS) was successfully applied to determine volatile compounds in 36 sesame-aroma baijiu samples and total of 125 volatile compounds were detected (Sun et al., 2018). However, due to the few pre-treatment procedures, the large amount of water retained in baijiu sample may damage the chromatographic column and detector. Additionally, the variety of compounds detected is also relatively limited due to no enrichment step in the sample processing process, so it is not suitable for the analysis of trace components with relatively low content.

Liquid-liquid extraction (LLE) is a method to add a selected solvent which is not miscible (or slightly miscible) to the ethanol-water solution, and then use the different solubility of each component in selected solvent to achieve separation or extraction. Most of the trace components in baijiu have a certain polarity, so the commonly used extraction solvents are dichloromethane, ether and so on. As reported, the extraction efficiency could be improved by adding salt into the sample and diluting the sample. Compared with DI, LLE has the advantages that it can concentrate and enrich the trace components. Furthermore, it has high recovery rate and wide application range. Thus, LLE is the most widely used pretreatment method at present. In 2013, Xu et al. used LLE to extract the volatile compounds from Fenjiu and Langjiu (Fan and Xu, 2013a, 2013b). As a result, 1500 peaks were detected and 698 kinds of volatile components were identified. In 2018, Wang et al. used LLE to extract the main trace components of Niulanshan light-aroma base baijiu (Wang et al., 2018). Consequently, a total of 109 compounds were confirmed, including 49 esters, 19 alcohols, 6 aldehydes, 5 ketones, 6 acids, 11 hydrocarbons, 8 terpenes and 5 phenolic compounds. However, the operation of this method is relatively complicated, and the volume of organic solvents required to be used is relatively large, thus, there will be some loss of samples in the extraction and enrichment process.

Table 2
Common detection technique and characteristics of trace components in baijiu.

Detection technique	Characteristics	Suitable conditions
Mass-spectrometry (MS)	Conventional detector with wide range of applications	All kinds of components
Flame ionization detector (FID)	High-sensitivity conventional-purpose detector with good stability and quick response	All kinds of components
Flame photometric detector (FPD)	High sensitivity and selectivity for sulfur components	Sulfur compounds
Nitrogen phosphorus detector (NPD)	High sensitivity and selectivity for nitrogen components	Nitrogen compounds, such as pyrazines
Sulfur chemiluminescence detector (SCD)	The most sensitive and selective detector for the detection of sulfur components	Sulfur components
Comprehensive two-dimensional gas chromatography (GC×GC)	Great peak capacity, good separation and sensitivity, and is suitable for the separation of complex mixtures	All kinds of components
Liquid chromatography (LC)	High efficiency, high sensitivity and good selectivity	Components with high boiling point, strong polarity and poor thermal stability
Gas chromatography-olfactometry (GC–O)	Has a high sensitivity for the detection of potential aroma components	Aroma active components

Compared with LLE, liquid-liquid microextraction (LLME) requires less sample consumption, shorter pretreatment time, and relatively simple steps. In 2012, Wang et al. applied LLME to extract the trace components of sauce-aroma baijiu. The results showed that 48 trace components were found, and 29 of them were confirmed as skeleton components (Wang et al., 2012). In 2020, Wang et al. used LLME to enrich phenolic compounds and fatty acids in Moutai Jiu, and successfully quantified the concentrations of 8 compounds with the limit of detection in the range of 7.0 × 10–10–8.7 × 10^{-4} g/L and the limit of quantification in the range of 2.3 × 10–9–1.7 × 10^{-3} g/L (Wang et al., 2020a). LLME has a certain enrichment ability, but the enrichment ability is slightly worse than LLE, so it is not suitable for the extraction of some low-content compounds.

Solid-phase extraction (SPE) is a method to separate trace components by utilizing the difference in the distribution balance between the stationary phase and the mobile phase. Due to its low consumption of sample and solvent, good universality and selectivity, SPE has been widely used. In 2007, Li et al. used SPE to enrich the trace components in baijiu, and the enriched samples were separated and identified by GC/MS (Li et al., 2007). As a consequence, a total of 50 components were verified from the neutral components, including 7 organic acids. In 2020, Wu et al. established a SPE-HPLC method for the qualitative and quantitative analysis of 5 phenolics in baijiu, including ferulic acid, gallic acid, p-coumaric acid, syringic acid and (+)-catechin hydrate(Wu et al., 2019). And these five phenolics were successfully determined in 46 baijiu samples. Compared with traditional extraction methods, SPE is easy to operate and it can effectively eliminate the matrix effect of high-concentration alcohol in baijiu samples. Besides, it has good repeatability in quantitative analysis. While the enrichment capacity of this method is also relatively weak.

Solid phase microextraction (SPME) is a derivative of SPE, which was proposed by Belardi and Pawliszyn in 1989 (Berladi and Pawliszyn, 1989). The most commonly used is headspace solid phase microextraction (HS-SPME). Different from SPE, this method is based on the equilibrium between the stationary phase and the gas phase, and without the need of organic solvents. Besides, HS-SPME is easy to automate online analysis. Hence, the method has a wide range of applications and can be used to analyze all kinds of volatile or semi-volatile components in gas, liquid and solid samples (Pu et al., 2019b). In 2017, Liu et al. analyzed the trace components of millet huangjiu by HS-SPME combined with GC–MS, and 7 kinds of characteristic volatiles were identified, including phenylethanol and benzaldehyde(Liu et al., 2017). In 2018, Deng et al. applied SPME-GC–MS to investigate the volatile compounds in baijiu. As a result, a total of 33 volatile compounds were confirmed, including 23 esters, 3 aldehydes, 2 acids and 5 ketones. They also proved that ester was the main volatile compounds in baijiu, and ethyl hexanoate has the highest content among all volatile compounds in mixed-aroma baijiu. Notably, the enrichment capacity of SPME is limited due to its thin coating, and the adsorption of weak volatile compounds is not good. Besides, the extraction fiber is relatively expensive.

Stir bar sorptive extraction (SBSE) is also a SPE sample pretreatment technique by adsorbing sample molecules via polymer coating. The stir bar is covered with an adsorption coating, and the specific compounds is adsorbed on the coating during stirring. The method has larger adsorption capacity than SPME, high sensitivity, selectivity and recovery rate, good reproducibility. In addition, SBSE can achieve extraction and enrichment while stirring by itself. In 2011, Fan et al. applied SBSE to extract the trace components of sauce-aroma baijiu. The results showed that a total of 76 volatile compounds were identified from 14 baijiu samples, including esters, alcohols, acids, phenols, aldehydes and

ketones, furans, sulfur compounds and nitrogen compounds (Fan et al., 2011). However, the non-volatile compounds will also be adsorbed on the stir bar, which will interfere with the analysis of semi-volatile compounds and volatile compounds.

Apart from the above methods, other extraction methods have also been applied to extract the volatile compounds in baijiu, such as simultaneous distillation extraction (SDE), microwave-assisted extraction (MAE) and purge and trap (PT) (Liu et al., 2010; Zhang, 2014). In 2015, Li et al. used the SDE to extract volatile compounds in Luzhou Laojiao of different ages (Li et al., 2015). The results showed that a total of 51 compounds were obtained. All the above extraction methods have its own characteristics and limitations, and have different selectivity for different trace components. Therefore, it is necessary to select one or more extraction methods according to the actual situation to ensure the comprehensive analysis on trace components of baijiu.

4. Detection technique of trace components in baijiu

With the development of science and technology, the detection technique of trace components in baijiu has been greatly improved. At present, mass spectrometry (MS), flame ionization detector (FID), flame photometric detector (FPD) and olfactory are widely used in the detection of trace components in baijiu (Fan et al., 2014; Jia et al., 2020; Xu et al., 2010). In addition, detectors are usually combined with gas chromatography because of its excellent separation ability on volatile compounds. The common detection techniques for analysis on trace components of baijiu was objectively summarized as shown in Table 2.

Mass-spectrometry (MS) is an analytical method which ionizes the compounds to be measured and separated according to the mass-charge ratio of ions, then monitors the strength of various ion spectrum peaks. Mass is one of the inherent characteristics of compounds, different compounds have different mass spectrum, which can be applied for qualitative analysis. The peak intensity is also related to the concentration of compounds, so it can be used for the quantitative analysis. Nowadays, gas chromatography-mass spectrometry (GC–MS) has been widely used in the study of trace components in baijiu. In 2017, direct injection combined with GC–MS was successfully applied to determine volatile compounds in 36 sesame-aroma baijiu samples and a total of 125 volatile compounds were detected (Sun et al., 2018).

Flame ionization detector (FID) is a high-sensitivity conventional-purpose detector. It has simple structure, good stability and quick response to almost all organic compounds. In 2012, DI coupled with GC–FID was applied to quickly separate and quantify 20 skeleton components such as alcohols and esters in strong-aroma baijiu (Xiang and Zhang, 2012). Afterwards, in 2021, Yu et al. analyzed the aroma compounds of two different fermentation technology of strong-aroma baijiu through GC–FID combined with odor activity value (OAV) (Yu et al., 2021). The results showed that the types of aroma compounds of mechanized and traditional fermentation technology were similar, but the sample obtained from two fermentation technology had their own unique flavor composition and characteristics.

There are many kinds of compounds with low content and low threshold in baijiu, such as sulfur compounds and nitrogen compounds (Sun et al., 2021). The response of these compounds on conventional detectors (i.e., MS and FID) are very low, and their peaks are often masked by the peaks of higher concentrations of compounds, thus, it is difficult for them to be identified by conventional detectors alone. Therefore, specific detectors have been developed for these low-content compounds.

Flame photometric detector (FPD) is a specific detector mainly used for the analysis of sulfur compounds in baijiu, with high sensitivity and selectivity. In 1994, Hu et al. applied GC–FPD to analyze the sulfur compounds in sesame-aroma baijiu, and 3-methylthio-propanol was identified as the characteristic compounds of sesame-aroma baijiu(Hu et al., 1994). In 2017, Niu et al. used LLE combined with GC–FPD to analyze the light-aroma baijiu, and found that dimethyltrisulfide and 3-mercaptohexyl acetate had important contributions to the light-aroma baijiu (Niu et al., 2017b). With the development of technology, researchers have developed a new type of PFD, the pulsed flame photometric detector (PFPD). In 2017, based on GC–PFPD, Sha et al. solved the problem of difficulty in detecting sulfur compounds with low content in baijiu (Sha et al., 2017). The results showed that 18 sulfur compounds were identified in baijiu, among which 3-methylthiophene, 1-hexanethiol and 1-heptanethiol were found in baijiu for the first time.

Nitrogen phosphorus detector (NPD) is a highly sensitive and selective detector, which has a high response to nitrogen compounds. At present, NPD has been widely used in the determination of nitrogen compounds in baijiu. Nitrogen compounds such as pyrazine have low content and low threshold in baijiu, and generally with a roasted aroma, which make great contribution to the flavor of baijiu. In 2014, Wang et al. used LLE combined with GC–MS and GC–NPD to analyze nitrogen compounds in Guojing sesame-aroma baijiu (Wang et al., 2014a). The results showed that a total of 31 kinds of nitrogen compounds were detected in Guojing sesame-aroma baijiu, and 23 of them were accurately characterized by comparison of standard substances.

Sulfur chemiluminescence detector (SCD) is the most sensitive and selective detector for the detection of sulfur compounds. SCD can almost produce a linear response to various sulfur compounds without obvious quenching effect and interference. Its minimum detection limit for sulfur compounds is 0.2×10^{-9}, which is 100 times lower than FPD. In 2019, sulfur compounds in laobaigan baijiu were analyzed by HS–SPME and two-dimensional gas chromatography coupled with sulfur chemiluminescence detection (GC×GC–SCD) (Song et al., 2019). The results showed that 12 sulfur compounds were identified and they were determined to be in a range from 0.77 ± 0.02 μg/L to 60.04 ± 2.32 μg/L according to be further quantitated by the GC× GC–SCD. Of note, as reported, GC×GC–SCD possessed a higher sensitivity for these sulfur compounds due to the LODs of it was lower than that of other detectors and its minimum LODs for sulfur compounds was 0.2 ng/L.

Since the 1990 s, comprehensive two-dimensional gas chromatography (GC×GC) has become a novel method for separation of different types of complex mixtures. GC×GC using two columns with different properties to further separate compounds that are not completely separated in the first dimension. It greatly improves peak capacity, resolution and sensitivity, and is suitable for the separation of complex mixtures (i.e., baijiu). At present, the comprehensive two-dimensional gas chromatography combined with the time-of-flight mass spectrometer (GC×GC/TOFMS) has been commonly used in the flavor analysis of baijiu. In 2017, Chen et al. used HS–SPME as pretreatment method combine with GC×GC–TOFMS detection technique to analyze the volatile components in sesame-aroma baijiu (Chen and Xu, 2017). A total of 1029 chromatographic peaks were detected, and 340 volatile compounds were identified by various qualitative methods.

Liquid chromatography, with liquid as mobile phase, has the advantages of high efficiency, high sensitivity and good selectivity. It can realize the analysis of most organic compounds, especially those with high boiling point, macromolecules, strong polarity and poor thermal stability (Jia et al., 2020). High-performance liquid chromatography (HPLC) can be used in combination with UV detectors, diode array detectors, MS detector and other detectors. In 2019, Wu et al. established a rapid, simple, and efficient method for the determination of five phenolic compounds in alcoholic fermentation materials and spent grains by HPLC combine with UV detector (Wu et al., 2019). The technique has high separation performance and can realize the separation of multiple monomer compounds. At present, it has been widely used in the analysis of non-volatile organic acids, sugar components, amino acids, vitamins, proteins, biogenic amines in baijiu. However, considering that volatile compounds (e.g., alcohol, acid, ester, aldehyde) are the main components of baijiu, the application of HPLC in baijiu detection is not as extensive as GC.

Olfactometry, which uses the human nose as a detector, has a high sensitivity for the detection of potential aroma compounds (Pu et al.,

2019a; Pu et al., 2020). Nowadays, gas chromatography- olfactometry (GC–O) has been widely used to screen the aroma compounds and study the contribution of aroma compounds to the aroma profile of baijiu. In 2013, Wang et al. analyzed the aroma compounds in Xijiu (a kind of sauce-aroma type of baijiu) by GC–O combined with GC–MS (Wang et al., 2013a). A total of 72 aroma compounds were detected, among which volatile organic acids, alcohols and esters were deemed as important aroma compounds in Xijiu. Compared with the traditional analysis technique, GC–O links compounds with aroma, thus, lays an important material foundation for the screening and evaluation of aroma compounds from thousands of trace components in baijiu (Chen et al., 2021). Nevertheless, this method is subject to subjective influence, and many compounds may have similar aroma characteristics, so this method can only be used as an auxiliary qualitative method.

5. Research progress of trace components in strong-aroma type of baijiu

The research on trace components in strong-aroma baijiu has been carried out for more than 50 years. In the early stage, it mainly focused on the comprehensive investigation of trace components with higher concentrations in strong-aroma baijiu. In the process of research, it was found that there are many trace components in strong-aroma baijiu, but the qualitative method is single and the detection technique is limited, thus, accuracy is difficult to guarantee.

In 1964, the former Ministry of Light Industry organized researchers to investigate the volatiles in Luzhou Laojiao, and the researchers preliminatively concluded that *Jiaochi* aroma is mainly provided from the aroma of ethyl hexanoate (Shen, 1998). Hence, ethyl hexanoate was identified as the main aroma compounds of strong-aroma baijiu. In 1995, GC–MS was used to preliminarily analyze the trace components of Wuliangye fresh baijiu and aged baijiu (Lian et al., 1995). The results showed that the characteristic aroma of aged baijiu was prominent, and the proportion of trace components in aged baijiu was more harmonious. Then, researchers conducted research on strong-aroma baijiu from different regions. In 2000, Fan et al. compared the trace components of Wuliangye and Yanghe Daqu, two representative of Sichuan and Jianghuai regions of strong-aroma baijiu in China (Fan and Xu, 2000). Through the analysis of acids, alcohols, esters, aldehydes, heterocyclic compounds, and volatile sulfur compounds, it was suggested that although both of Wuliangye and Yanghe Daqu were strong-aroma *Daqu* baijiu, there were obvious differences in trace components and flavor characteristics between them. Compared with Yanghe Daqu, Wuliangye has higher total acid and total aldehyde content, lower alcohol and total ester content. Thus, Yanghe Daqu has a relatively sweeter taste than Wuliangye. With the improvement of detection technique and the enrichment of qualitative methods, more and more kinds of trace components have been verified.

In 2006, LLE coupled with normal chromatography was used to extract and separate volatile compounds in strong-aroma baijiu (Fan and Qian, 2006a). Then a total of 92 compounds were detected by GC–MS with double column (polar and non-polar columns). In 2013, Wang et al. applied 2 pretreatment methods (HS-SPME and LLME) followed by GC-MS to analyze the volatile compounds of 13 kinds of strong-aroma Xijiu samples (Wang et al., 2013b). The results showed 188 volatile compounds were identified by comparison with authentic standards, NIST 11 library searching, RI, and aroma characteristics. In 2015, a method of SBSE-Thermal Desorption combined with GC-MS was established, and a total of 74 volatile compounds were identified in 4 kinds of strong-aroma baijiu by comparison with previously reported RI and NIST 05 library searching (Niu et al., 2015). Among them, 36 kinds of ester compounds were identified, with the most number. In 2016, the trace components in Gujinggong jiu were characterized by liquid-liquid extraction coupled with GC–MS and 188 volatile compounds were identified by comparison with authentic standards, NIST 11 library searching and RI, including 74 esters, 26 alcohols, 25 aromatic compounds, 20 acids, 14 hydrocarbons, 8 aldehydes and ketones, 8 sulfur-containing compounds, 7 acetals, 4 heterocyclic compounds and 2 lactones (Li et al., 2016). However, the trace components in baijiu were very complex, and the separation degree and peak capacity of one-dimensional gas chromatography were limited, which cannot completely meet the requirements for the separation of trace components in baijiu, thus affecting the identification of trace components in baijiu.

Due to the limited capacity of one-dimensional gas chromatography system, two-dimensional gas chromatography was used to study the trace components of strong-aroma baijiu. In addition, the application of multi-dimensional gas chromatography has greatly increased the number of compounds detected. In 2007, the volatile compounds of strong-aroma baijiu were first detected with a full-two-dimensional-time-of-flight mass spectrometer (GC×GC–TOFMS) (Ji et al., 2007). The results showed that a total of 674 peaks were obtained and 342 trace components were identified by MS. In 2015, LLE and SPE combined with GC×GC/TOF–MS were used to analyze the volatile compounds of Luzhou Laojiao, and more than 1300 chromatographic peaks were preliminarily identified in 18 types of Luzhou Laojiao samples (Yao et al., 2015). Furthermore, in 2020, Song et al. accurately identified 350 trace components in 58 kinds of strong-aroma baijiu samples by LLE combining with GC×GC/TOF-MS (Song et al., 2020). All in all, 861 kinds of trace components have been found in strong-aroma baijiu, including 248 esters, 122 alcohols, 65 acids, 83 aldehydes, 71 nitrogen compounds and so on. Among them, esters, acids and alcohols account for 50% of the total, which are the skeleton components of strong-aroma baijiu.

Each baijiu sample has its own flavor characteristics. It is speculated that the difference in flavor characteristics of different baijiu is mainly caused by two aspects: one is the type of trace components, and the other is the concentration of trace components. Therefore, it is necessary to determine the concentration distribution of volatile compounds after having a clear understanding of the types of trace components in strong-aroma baijiu. Based on qualitative analysis, the quantitative analysis of volatile compounds is essential for the determination of the volatiles in baijiu and further investigation on the contribution of the aroma compounds on the aroma profile of baijiu.

In 1995, Lian et al. analyzed the trace components of Wuliangye fresh baijiu and aged baijiu by GC–MS (Lian et al., 1995). As the results of quantification, the researchers found that the aged baijiu had high concentration of senior alcohols, which was speculated to have a great influence on the flavor of strong-aroma *Daqu* baijiu. In 1996, the trace components in Luzhou Laojiao Daqu jiu were quantified, and the concentration s of trace components and the ratio between them were regarded as having a great impact on flavor quality of baijiu (Huang et al., 1996). That was within a certain concentration range, the higher the concentration of acids and esters, the better the quality of baijiu. Subsequently, in 2007, Fan et al. used HS-SPME combined with gas-flame thermal ion detector (GC–FTD) to analyze strong-aroma baijiu (FTD is also a specific detector for nitrogen compounds) (Fan et al., 2007). And seven kinds of pyrazine compounds, including 2,5-dimethylpyrazine, 2,6-dimethylpyrazine and 2,3,5-trimethylpyrazine, were detected in five kinds of baijiu, such as Gujinggong jiu, Yanghe Daqu and Wuliangye.

Normalization method is a simple and fast quantitative method used in the quantification of trace components in baijiu. In 2009, trace components in strong-aroma baijiu were quantified by GC-MS coupled with area normalization method (Wu et al., 2009). The results showed that a total of 78 compounds were detected. Further analysis exhibited that the peak area of esters, acids and alcohols accounted for more than 90% of all trace components, which confirmed the main compounds composition of strong-aroma baijiu. Of note, because the response values of different compounds in the detector are different, the accuracy of normalization method is poor.

In 2007, DI combined with GC–FID and external standard

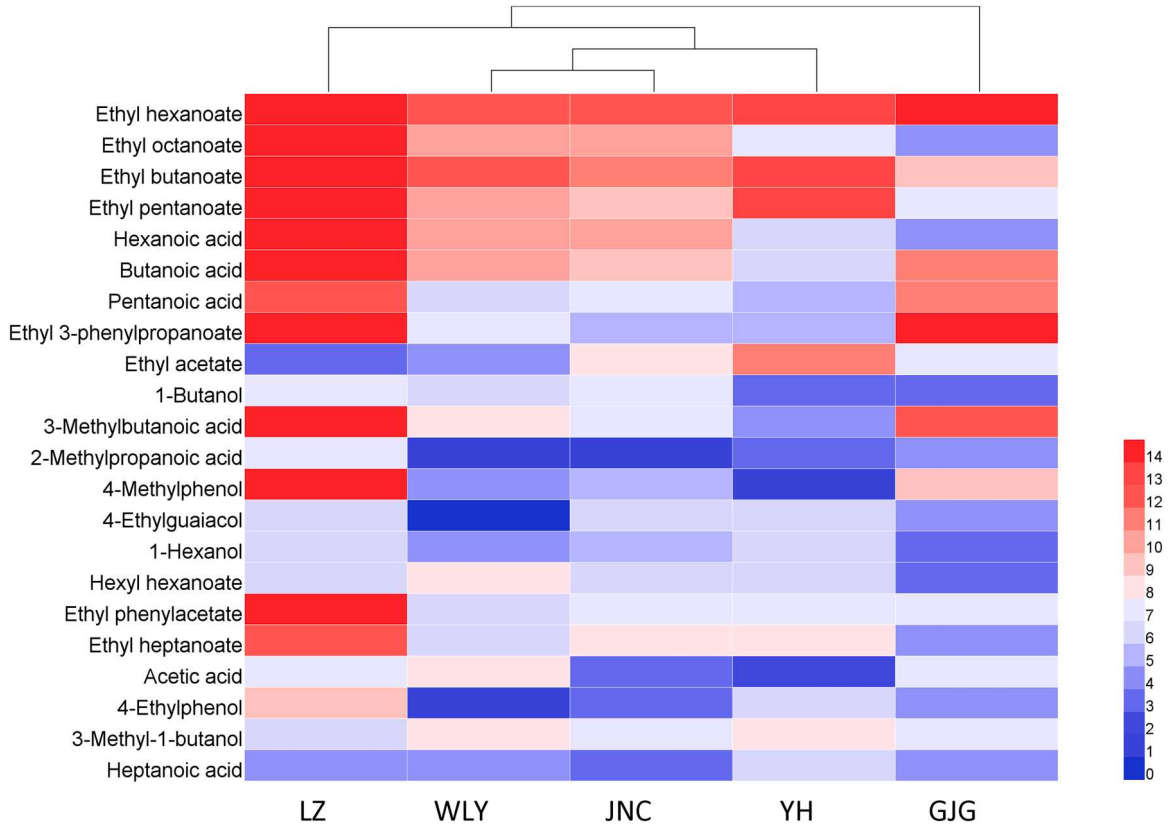

Fig. 7. A clustered heatmap for aroma compounds with high FD factors in typical brand of strong-aroma baijiu. The vertical labels show aroma compounds, and the horizontal labels represent different brand of strong-aroma baijiu.

quantitative method were applied to determine the concentrations of trace components in fresh baijiu of Sichuan region and Jianghuai region (Fan and Xu, 2007). The results showed that the total amount of trace components of strong-aroma baijiu produced in Sichuan area was 29 ~ 88 g/L, in which the total esters were up to 16 ~ 57 g/L and that in Jianghuai region was 10.61 g/L, in which the total esters were up to 7 ~ 9 g/L. From the point of view of concentration for trace components, the fresh baijiu in Sichuan region and Jianghuai region have great variance. In 2014, a GC-MS with external standard quantitative method was established for the determination of ethyl acetate, an important aroma compounds in baijiu (Huang, 2014). The results were within the error range, and the method could be used in daily analysis of ethyl acetate. In general, the external standard quantification method can calibrate the response value of the detector. But the accuracy of this method is influenced by the accuracy of injection volume and the stability of instrument.

Compared with the external standard method, the internal standard method does not strictly require the injection volume for quantification. The addition of the internal standard can eliminate the systematic error caused by the injection volume and instrument instability, thereby, enhancing the accuracy of results. Therefore, the internal standard method is widely used in the analysis of trace components in baijiu. In 2012, Xiang et al. applied direct injection and GC–FID combined with internal standard method to rapidly separate and quantify volatile compounds in strong-aroma baijiu, and 20 kinds of skeleton components such as alcohols and esters were identified and quantified (Xiang and Zhang, 2012). Besides, SIM is a commonly used quantitative mode based on GC–MS, which can significantly reduce signal noise and improve sensitivity and specificity by scanning only selected ions of target compounds. In 2015, propyl lactate in baijiu was analyzed by GC–MS/SIM coupled with internal standard quantitative method (Wu et al., 2015). The results showed that propyl lactate was detected in both Wuliangye and Gujinggong jiu, and the contents were 0.189 $\pm$ 0.004 mg/ L and 0.314 $\pm$ 0.002 mg/ L, respectively. In 2019, Dong et al. used various quantitative methods to quantify aroma compounds with high FD values in strong-aroma baijiu (Dong et al., 2019). The internal standard quantitative method was able to determine all the aroma-active compounds in the tested samples; moreover, this method had good linearities with correlation coefficients (R^2 values) ranging from 0.9907 to 0.9998 and RSDs $\leq$ 10% (n = 3). In the same year, Xu applied the pre-treatment method of LLE combined with GC–MS to analyze Zhijiang base baijiu samples, and 132 trace components were identified and quantified by internal standard quantitative method (Xu, 2019). Among them, the content of esters, fatty acids, alcohols, and carbonyl compounds was relatively high, accounting for 69.26%, 14.42%, 9.43% and 5.75% of the overall aroma content, respectively.

Stable isotope dilution is a novel quantitative method with high accuracy because it can eliminate both instrument systematic error and random error, and reduces the influence of matrix effect. Nowadays, it has been applied in quantitative analysis on volatile compounds in alcoholic beverage. Based on the previous qualitative results, Yanping L. Qian et al. analyzed the contents of volatile compounds in Qingke liquor by means of stable isotope dilution assay (Qian et al., 2019). The results showed that 40 aroma compounds had FD factors $\geq$ 64 and all the compounds had concentrations higher than their threshold except propyl acetate and α-terpineol in Qingke liquor samples, which indicate that they were vital aroma compounds. But isotopes need to be synthesized, and the process is complicated and expensive, so the application of this method is limited. The current quantitative results show that the compounds with larger concentration in strong-aroma baijiu are esters, alcohols, and acids. While the number of nitrogen compounds is also large but the concentration is generally low.

However, the researchers found that not all the trace components detected in strong-aroma baijiu have a direct contribution to the overall

aroma of strong-aroma baijiu. Besides, the low concentration of trace components does not mean that the contribution to the aroma characteristics of strong-aroma baijiu is not large, so it is necessary to screen aroma compounds. Therefore, GC–O analysis was introduced to explore and evaluate the potential aroma compounds from thousands of trace components in strong-aroma baijiu. According to different principles, GC–O analysis can be roughly divided into the following three types, time intensity analysis, dilution analysis and frequency analysis (Linssen et al., 1993; Ullrich and Grosch, 1987; Zellner et al., 2008). Among them, Osme and AEDA are more commonly used.

Time intensity analysis (e.g., Osme), is a method that evaluates the contribution of aroma compounds based on their odor intensity. Osme spectrum is obtained by recording the change of odor intensity over time and corresponding odor characteristics. Thus, it is widely used in the evaluation of aroma compounds in baijiu because of its low requirement on evaluators. In the Osme spectrum, the higher the peak is deemed as the greater the odor intensity of the compound and the greater the contribution to the overall aroma. In 2006, researchers applied GC–O combined with Osme to explore the potential aroma compounds of Yanghe Daqu and to evaluate the aroma expression intensity (Fan and Qian, 2006a). The results clarified that esters (such as ethyl hexanoate, ethyl butanoate) and fatty acid compounds (hexanoic acid, butanoic acid, 3-methylbutanoic acid, etc.) had higher Osme values, which gave baijiu prominent fruity and sour characteristics, respectively. Afterwards, Du et al. discovered and evaluated geosmin, an earth-smelling compound in baijiu by GC–O combined with Osme (Du et al., 2011). However, the aroma expression intensity of aroma compounds observed by evaluators is subjective and poor reproducible. Hence, other evaluation methods are gradually introduced.

Dilution analysis has become a widely used evaluation method based on GC–O due to its simple operation, among which AEDA is the most widely used method. AEDA was proposed in 1987, in which the contribution of each aroma compound was represented by FD factor (the maximum dilution factor of the sample required to be smelted by at least one sensory evaluator) (Ullrich and Grosch, 1987). It is generally believed that the larger FD factor, the greater its role in the presentation of matrix aroma characteristics. Compared with Osme, AEDA is supposed to be less subjective and more stable. In 2005, a total of 85 aroma compounds were perceived by GC–O in Yanghe Daqu jiu and further evaluated by AEDA, among which the aroma compounds with high FD factor were ethyl hexanoate, ethyl butyrate, ethyl pentanoate, ethyl heptanoate, butyl hexanoate, acetaldehyde, acetal and so on (Fan and Qian, 2005). In 2006, GC–O–AEDA was applied to analyze aroma compounds in Jiannanchun and Wuliangye, the two other strong-aroma baijiu representatives (Fan and Qian, 2006b). Consequently, ethyl butyrate, ethyl pentanoate, ethyl hexanoate, butyl hexanoate, ethyl 3-methylbutyrate and hexanoic acid contributed significantly to the aroma characteristics of strong-aroma baijiu. Among them, esters, especially ethyl ester compounds were the most important aroma compounds. In 2017, AEDA–GC–O was applied to analyze aroma compounds in Wuliangye jiu, and 62 aroma compounds were identified in three different aged Wuliangye jiu, of which 45 compounds have FD values ≥ 16 (Niu et al., 2017a). Through correlation analysis, it was found that 11 compounds such as ethyl hexanoate, furfural, 4-methylphenol and acetal were positively correlated with the characteristic aroma of Wuliangye. To sum up, the skeleton aroma compounds in five typical strong-aroma baijiu were collected to draw the heatmap (as shown in Fig. 7) according to their FD factors. The gradient of color from red to blue represents the FD factors of target aroma compounds from the highest to the lowest. According to the clustering results, it can be seen that Wuliangye and Jiannanchun had a high similarity, which was consistent with that they were both Sichuan region multi-grain strong-aroma baijiu. Compared with Jianghuai region strong-aroma baijiu (Yanghe Daqu and Gujinggong jiu), the Sichuan region strong-aroma baijiu (Luzhou Laojiao) has a higher content of main esters and acids, which may be the reason for the difference in flavor between the two types strong-aroma baijiu.

With the further development of analytical technique, researchers have found that the aroma smelled based on GC–O only represents the odor intensity of the aroma compound in the air matrix, and cannot indicate the contribution of this aroma compound in the complex matrix of baijiu. Besides, the concentration of aroma compounds in baijiu could not directly reflect the contribution of aroma compounds to the aroma profile of baijiu. Certain low-content sulfur, nitrogen and furan compounds in baijiu also make significant contributions to the overall aroma. In order to further evaluate their flavor contribution, the concepts of odor threshold (the minimum concentration of an aroma compound at which people can smell the aroma) and odor activity value (OAV, the ratio of the content of the compound to the odor threshold) were proposed (Acree et al., 1984). The appearance of OAV has greatly promoted the evaluation of characteristic aroma compounds in baijiu. Only aroma compounds with OAV above 1 is deemed to have a contribution to baijiu flavor. And it is generally acknowledged that the higher the OAVs of aroma compounds, the greater the contribution to the overall flavor of baijiu.

In 2011, according to the method of GB T33406–2016, Fan et al. determined the odor threshold of 79 representative aroma compounds in baijiu, such as ethyl hexanoate, ethyl acetate and ethyl lactate (Fan and Xu, 2011). This study promoted the application of OAV in the evaluation of aroma compounds in baijiu. In 2013, the term system of strong-aroma baijiu flavor wheel was established by studying the establishment methods of foreign liquor products and combining the characteristics of baijiu and the identification results of aroma compounds in baijiu (Zhou et al., 2013). In addition, 19 flavor description language were defined to unify the sensory description standard of baijiu. Based on the above study, in 2014, Wang et al. extracted aroma compounds in Xijiu by LLE (Wang et al., 2014b). Meanwhile, the results of GC–MS–O showed that a total of 68 aroma compounds were detected in Xijiu, and 22 aroma compounds with OAVs > 10, including ethyl hexanoate, ethyl octanoate, butanoic acid, 3-methylbutanoic acid, hexanoic acid, dimethyl disulfide and so on. Due to their higher OAVs, these aroma compounds were considered having great contribution to the whole aroma profile of Xijiu. In 2019, in order to clarify the characteristic aroma compounds of the strong-aroma baijiu of Yimeng old district, Zhang et al. used LLE combined with GC-O to analyze the original and finished baijiu of Yimeng old district (Zhang, 2019). And a total of 56 aroma compounds were identified by GC-O and GC-MS. According to the OAVs, 15 aroma compounds, such as ethyl hexanoate, hexanoic acid and butyric acid, were considered as important aroma compounds in strong-aroma baijiu from Yimeng old district.

The above mentioned AEDA and OAV both take a single compound as the object to evaluate their respective aroma expression, and do not consider the actual interaction between aroma compounds in baijiu matrix. In order to further evaluate the contribution of aroma compounds to the overall flavor of baijiu, sensory evaluation experiments (e.g., aroma recombination and omission analysis) have been carried out gradually. Aroma recombination is based on AEDA and OAV to select target compounds with higher FD factors or OAVs for the recombination of target sample. Then evaluate the contribution of aroma compounds to the overall flavor of baijiu by comparing the recombination samples with real baijiu samples. The omission experiment is based on the recombination experiment to further confirm the key aroma compounds that have significant impacts on the aroma profile of baijiu.

In 2018, Zhao et al. used DI, LLE, SPME as pretreatment methods, and combined with GC–O and GC–MS to extract and analyze volatile compounds in Gujinggong jiu (Zhao et al., 2018). Consequently, 60 aroma compounds were identified, among which, 35 aroma compounds were further recognized as the important aroma compounds because of their relatively high OAVs. Furthermore, these 35 aroma compounds were recombined according to its concentration in Gujinggong jiu and the aroma of the simulation sample was similar to the real baijiu sample. Of note, omission experiment also proved that esters and acids played

Table 3
Main aroma compounds in strong-aroma baijiu.

No.	Category	Compounds	CAS Number	Odor thresholds (μg/L)	Descriptor[a]	OAV				FD factor			
						ZJ[b]	GJG[c]	LZLJ[d]	YH[e]	GJG[c]	YH[e]	WLY[f]	JNC[f]
1	Esters	Ethyl pentanoate	539–82–2	26.78[g]	Fruity, pineapple	308.81	1316	5369	30.19	243	> 8192	4096	512
2		Ethyl acetate	141–78–6	32,551.60[g]	Pineapple	44.19	40	54	105.46	243	2048	256	256
3		Ethyl heptanoate	106–30–9	13,200.00[g]	Fruity	2.64	4	3	61.4	27	256	32	256
4		Ethyl butanoate	105–54–4	81.50[g]	Pineapple	5516.99	2024	7634	5713.34	729	> 8192	4096	
5		Ethyl propanoate	105–37–3	19,019.33[g]	Fruity	0.59	< 1		5.64	243	8		
6		Ethyl hexanoate	123–66–0	55.33[g]	Pineapple	60,701.43	36,848	137,558	403,100	59,049	> 8192	4096	
7		Propyl hexanoate	626–77–7	12,800[g]	Fruity	0.14	< 1	1		81		16	
8		Butyl hexanoate	626–82–4	-[h]	Fruity						1024	256	
9		Pentyl acetate	628–63–7	–	Fruity						64		
10		Ethyl 2-methylbutyrate	7452–79–1	–	Fruity	1030		35			256	32	64
11		Ethyl 3-methylbutyrate	108–64–5	6.89[g]	Fruity, apple, pineapple	274.17		264	166.11		128	512	1024
12		Pentanoic acid,4-methyl-, methyl ester	2412–80–8	–	Fruity						256		
13		Pentanoic acid,4-methyl-, ethyl ester	25415–67–2	6[i]	Fruity			155			128		
14		3-methylbutyl 3-methylbutanoate	659–70–1	19,807.17[g]	Fruity						16		
15		3-Methylbutyl butanoate	106–27–4	–	Fruity						32		
16		2-Methyl-1-propyl butyrate	539–90–2	–	Fruity						8		
17		Octanoic acid, 3-methylbutyl ester	2035–99–6	–	Fruity, pineapple				3.69		8		
18		Butyl acetate	123–86–4	–	Fruity				0.01		1		
19		Methyl hexanoate	106–70–7	–	Fruity							128	32
20		Ethyl nonanoate	123–29–5	–	Waxy	8.07			0.14	9	32	64	
21		Ethyl lactate	97–64–3	128,000[g]	Fruity	17.67	7		7.05	27			
22		Ethyl octanoate	106–32–1	12.87[g]	Banana	8512.04	5303	3401	17,734.49	27		512	1024
23		Propyl octanoate	624–13–5	–	Coconut						32		
24		Hexyl butyrate	2639–63–6	8620[i]	Green			< 1			8		
25		Pentyl hexanoate	540–07–8	–	Fruity							4	
26		Hexyl hexanoate	6378–65–0	1890[i]	Apple, peach		9	15			64	256	128
27		Heptyl acetate	112–06–1	–	Fruity						4		
28		Pentyl butanoate	540–18–1	–	Fruity						2		
29		Ethyl 2-methylpropanoate	97–62–1	57.47[i]	Sweet, fruity	75.41		166			8		
30		2-Hydroxy-3-methylbutanoic acid ethyl ester	2441–06–7	–	Fruity						8	32	8
31		Isobutyl acetate	110–19–0	–	Fruity						2		
32		Isoamyl acetate	123–92–2	93.93[g]	Banana	62.94			33.12				
33		Isobutyl hexanoate	105–79–3	5250[i]	Apple, fruity		1	< 1		9		32	64
34		Diethyl succinate	123–25–1	353,193.25[g]	Fruity	0.02	< 1	< 1		9	16		
35		Ethyl decanoate	110–38–3	1122.30[g]	Green, fruity	2.58		1	4.98		64	16	32
36		Hexyl octanoate	1117–55–1	–	Green, herb						32		
37		Hexyl acetate	142–92–7	–	Apple, banana	1.42			0.65		8		
38		Isopentyl hexanoate	2198–61–0	–	Fruity		5			27		256	
39		Ethyl cyclohexanecarboxylate	3289–28–9	–	Fruity, floral							256	256
40		Ethyl dodecanoate	106–33–2	–	Sweet, waxy	0.01	2			81		8	8
41		Ethyl tetradecanoate	124–06–1	–	Waxy					9		1024	
42		Ethyl hexadecanoate	628–97–7	–	Waxy					9		64	
43		Ethyl oleate	111–62–6	–	Fatty					3			
44		Ethyl linoleate	544–35–4	–	Fatty					3			
45		Ethyl 2-hydroxybutanoate	52089–54–0	–	Green					3			
46		2-Hydroxyhexanoic acid ethyl ester	52089–55–1	–	Fruity						32		
47	Acids	Acetic acid	64–19–7	160,000[i]	Vinegar	2.78	3	5		243	4	32	8
48		Propionic acid	79–09–4	18,100[i]	Sweaty, acidic	0.59	3	1				16	8
49		Butanoic Acid	107–92–6	964.64[g]	Rancid, cheesy	380.22	209	307		2187	64	512	512
50		Hexanoic acid	142–62–1	2517.16[g]	Sweaty, cheesy	143.84	316	447		19,683	64	1024	1024

(*continued on next page*)

Table 3 (*continued*)

No.	Category	Compounds	CAS Number	Odor thresholds (μg/L)	Descriptor[a]	OAV				FD factor			
						ZJ[b]	GJG[c]	LZLJ[d]	YH[e]	GJG[c]	YH[e]	WLY[f]	JNC[f]
51		Pentanoic acid	109–52–4	389.11[g]	Sweaty, cheesy	49.96	89	146		2187	32	256	128
52		Heptanoic acid	111–14–8	13,300.00[g]	Sweaty	0.59	1	2		27	64	1024	8
53		Nonanoic acid	112–05–0	3559.23[g]	Fatty		< 1			3			
54		Octanoic acid	124–07–2	2701.23[g]	Sweaty, cheesy	5.44		10				16	8
55		Isobutyric acid	79–31–2	1580[i]	Cheesy, acidic			11			8	16	2
56		4-Methylpentanoic acid	646–07–1	144[j]	Sweaty		12	9		27		16	8
57		3-Methylbutanoic acid	503–74–2	1045.47[g]	Cheesy, acidic	27.03	18	13		6561	16	256	128
58	Alcohols	1-Butanol	71–36–3	2733.35[g]	Alcoholic	32.21	40	54	0.23	9	8	64	128
59		2-Butanol	78–92–2	–	Fruity						32	1	
60		2-Pentanol	6032–29–7	–	Alcoholic, fruity						256	8	
61		1-Pentanol	71–41–0	–	Fruity	1.49						128	128
62		1-Hexanol	111–27–3	5370[i]	Green	6.71	11	18	4.91	9	64	1024	32
63		1-Propanol	71–23–8	53,925.63[g]	Alcoholic	3.77			1.14				
64		3-Methyl-1-butanol	123–51–3	179,000[g]	Fruity	2.14	1		4.82	243	256	128	32
65		2-Methylbutan-1-ol	137–32–6	179.19[i]	Fruity					243			
66		1-Octanol	111–87–5	1100[j]	Waxy, green		1	2	0.45		8		
67		2-Ethylhexanol	104–76–7	–	Green, rose							8	16
68		2-Methyl-propanol	78–83–1	40[j]	Fruity, apple		4		1.1	9		4	1
69		1-Heptanol	111–70–6	–	Green				109.86		8		
70		2-Heptanol	6033–23–4	1433.94[g]	Mushroom		2			9	2	2	2
71	Aromatic compounds	2-Phenylethyl acetate	103–45–7	407.00[g]	Honey, rose		< 1		0.94	9	128	32	64
72		Ethyl 3-phenylpropionate	2021–28–5	125[g]	Flower, honey	54.32	58	13	719.54	19,683	32	128	32
73		Ethyl phenylacetate	101–97–3	909[g]	Floral, honey, rose	10.72	6	5		243		64	128
74		Ethyl 3-phenylacrylate	103–36–6	–	Floral, honey	20							
75		Hexanoic acid, 2-phenylethyl ester	6290–37–5	908.83[g]	Fruity						4	64	
76		Ethyl benzoate	93–89–0	1430.00[g]	Floral	0.87		< 1	1.04		64	256	
77		Phenylacetic acid	103–82–2	1430[g]	Honey, floral			< 1					
78		3-Phenyl propionic acid	501–52–0	420[g]	Floral			1		81			
79		Benzyl alcohol	100–51–6	40,927.16[g]	Floral	0.06							
80		Phenethyl alcohol	60–12–8	28,900[g]	Rosy	0.17	< 1	< 1	0.06	243	2		128
81		Phenylacetaldehyde	122–78–1	262[g]	Floral, clove	5286	2	7		81			
82		Benzaldehyde	100–52–7	2000[g]	Almond, fruity	0.58							
83		4-Ethylguaiacol	2785–89–9	122.74[g]	Clove	1.95	12		2.06	27	64		64
84		2,4-Di-tert-butylphenol	96–76–4	–	Phenol					9			
85		Guaiacol	90–05–1	13.41[g]	Phenolic, smoky	11.19							
86		Phenol	108–95–2	18,909.34[g]	Phenol, medicinal	0.06	< 1		0.04	27			32
87		4-Methylphenol	106–44–5	167[g]	Animal		15	20	0.17	729	2		32
88		4-Ethylphenol	123–07–9	618[g]	Smoky	0.05	2	< 1	0.13	27	64		8
89		4-Methylguaiacol	93–51–6	314.56[g]	Smoky		3	1		81			
90		4-Vinylguaiacol	7786–61–0	209.3[g]	Clove			< 1					
91		Ethyl 2-hydroxy-3-phenylpropanoate	15399–05–0	–	Smoky							16	4
92		Vanillin	121–33–5	439[j]	Vanilla, sweet		< 1	< 1		729			
93	Aldehydes and ketones	Acetaldehyde	75–07–0	–	Green	0.07					256		
94		2-Methylpropanal	78–84–2	–	Green						64	2	64
95		Pentanal	110–62–3	–	Fermented						1		
96		Butyraldehyde	123–72–8	2901.87[g]	Fruity, floral	0.16							
97		2-Methylbutyraldehyde	96–17–3	–	Green						2		
98		3-Methylbutyraldehyde	590–86–3	–	Green, malty						16	16	8
99		Hexanal	66–25–1	25.48[g]	Apple, green grass							256	8
100		Octanal	124–13–0	39.64[g]	Green, herbal		8			243			
101		Nonanal	124–19–6	122.45[g]	Soapy	3.36	3			9		16	
102		2-Octanone	111–13–7	50[j]	Earthy				11.44				
103		2-Pentanone	107–87–9	–	Fruity						4		
104	Furans	Furfuryl acetate	623–17–6	26,050.00[g]	Caramel, sweet			< 1				16	8

(*continued on next page*)

Table 3 (*continued*)

No.	Category	Compounds	CAS Number	Odor thresholds (μg/L)	Descriptor[a]	OAV				FD factor			
						ZJ[b]	GJG[c]	LZLJ[d]	YH[e]	GJG[c]	YH[e]	WLY[f]	JNC[f]
105		2-Furancarboxylicacid, ethyl ester	614–99–3	132,000.00[g]	Balsamic			< 1				8	8
106		Furfuryl butanoate	623–21–2	–	Sweet, caramel							64	8
107		2-Acetylfuran	1192–62–7	58,504.19[g]	Sweet, caramel							2	32
108		Furfural	98–01–1	44,029.73[k]	Sweet, almond	1.41			1.05		32	128	32
109		5-Methyl furfural	620–02–0	466,321.08[g]	Caramel	0.01						4	8
110		Hexanoic acid,2-furanylmethyl ester	39252–02–3	–	Fruity						32	1	8
111		1-(2-Furanyl)-ethanone	110–00–9	–	Malty					9			
112		3-Methyl-2(5 H)-furanone	22122–36–7	–	Cooked, roasted		< 1			9			
113		2-Acetyl-5-methylfuran	1193–79–9	40,870.06[j]	Nutty	0.01						4	
114		2-Butylfuran	4466–24–4	–	Spicy							64	
115		2-Furanmethanol	98–00–0	–	Burnt sugar							64	32
116	Pyrazines	2-Methylpyrazine	109–08–0	121,927.01[k]	Nutty	0.01							
117		2,5-Dimethyl pyrazine	123–32–0	3201.9[g]	Roasted	0.43							
118		2,6-Dimethylpyrazine	108–50–9	790.79[j]	Cocoa	0.43							
119		2,3,5-Trimethylpyrazine	14667–55–1	729.86[g]	Nutty	1.42	< 1			81			
120		2,3,5,6-Tetramethyl pyrazine	1124–11–4	80,073.16[g]	Nutty	0.13	< 1			27			
121		2-Ethyl-6-methylpyrazine	13925–03–6	–	Nutty, roasted							128	4
122		2,5-dimethyl-3-ethylpyrazine	13360–65–1	–	Roasted, baked							256	8
123	Sulfur and nitrogen	Dimethyl disulfide	624–92–0	9.13[g]	Cabbage						8	16	8
124	compounds	Dimethyl trisulfide	3658–80–8	0.36[g]	Cooked onion	416.67		467			32	32	2
125		Ethyl 3-methylthiopropionate	13327–56–5	3080[j]	Sulfureous			< 1					
126		3-(Methylthio)propanol	505–10–2	2110.14[g]	Meaty			< 1					
127		Methional	3268–49–3	7.12[i]	Cooked potato			52					
128		3-Methylindole	83–34–1	6.09[i]	Mud			8					
129	Acetals	1,1-Diethoxy-2-methylbutane	3658–94–4	–	Fruity						8	32	8
130		1,1-Diethoxy-3-methylbutane	3842–03–3	–	Fruity					9	2048	4096	1024
131		1,1,3-Triethoxypropane	7789–92–6	3700[j]	Vegetal		< 1			3	16	16	2
132		Nonanaldiethylacetal	54815–13–3	–	Fruity, floral						64		
133		(2,2-Diethoxyethyl)-benzene	6314–97–2	13,186.43[k]	Fruity					81	32	8	64
134		Acetaldehyde	105–57–7	2090[k]	Creamy, fruity	413.65					2048	256	256
135		1,1-Diethoxy-2-methylpropane	1741–41–9	–	Fruity						2	2	32
136		1,1-Diethoxyhexane	3658–93–3	–	Floral							64	8
137	Lactones	γ-Octalactone	104–50–7	2816.33[g]	Coconut, Fruit	0.01						32	2
138		γ-Decalactone	706–14–9	10.87[g]	Fruity	4.6							
139		γ-Hexalactone	695–06–7	–	Sweet		< 1			81			
140		γ-Nonanolactone	104–61–0	90.66[g]	Sweet, creamy, coconut	7.39	3	1		2187		8	64
141		γ-Heptalactone	105–21–5	–	Sweet, coconut		< 1			81			

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[a] The descriptor of aroma compounds referred to www.thegoodscentscompany.com.

[b] Odor thresholds was taken from reference (Xu, 2019).

[c] Odor thresholds was taken from reference (Zhao et al., 2018).

[d] Odor thresholds was taken from reference (Dong et al., 2019).

[e] Date was taken from reference (Fan and Qian, 2006a).

[f] Date was taken from reference (Fan and Qian, 2006b).

key roles in the aroma profile of strong-aroma baijiu. Besides, γ-nonalactone, 4-methylphenol, and ethyl 3-phenylpropanoate were also confirmed as the novel key aroma compounds of Gujinggong jiu. In 2019, two characteristic aromas (the roasted and mud-like aromas) of strong-aroma base baijiu and the interactions among the corresponding key compounds causing these two aromas were investigated by GC–MS and AEDA–GC–O (Dong et al., 2019). A total of 68 odorants were identified in Luzhoulaojiao base baijiu by comparative aroma extract dilution analysis and 37 odorants were shown to be important odorants as their OAVs ≥ 1. More importantly, based on aroma recombination and omission analysis, hexanoic acid, butanoic acid, p-cresol, 3-methylindole, and 3-(methylthio)propanal were confirmed to be the crucial aroma compounds responsible for the mud-like and roasted odors in strong-aroma base baijiu. It was worth noting that 3-methylindole was detected in strong-aroma baijiu for the first time.

Moreover, the typical sensory style of strong-aroma baijiu is the result of the combined action of various aroma compounds contained in it. When the aroma compounds are mixed with each other, the aroma molecules will react with each other due to physical and chemical reactions, and then the aroma of the liquor body will have subtle changes. Therefore, the aroma characteristics of baijiu are not equivalent to the simple addition of all the single aroma compounds contained in it. Investigating the interaction between aroma compounds is very crucial to further analyze and clarify the complex aroma characteristics of strong-aroma baijiu.

In recent years, there have been many researches on the interaction between aroma compounds in beer, red wine, whiskey, and other alcohol products, while the research on baijiu is relatively few. In 2018, Niu et al. used odor thresholds to investigate the perceptual interaction of 31 binary mixtures composed of 18 important esters in light-aroma baijiu (Niu et al., 2018). The results showed that compounds with similar structures and odors mainly presented either a synergistic effect or additive action. Hence, strengthening the research on the interaction between the aroma compounds of strong-aroma baijiu will provide an important theoretical basis for improving the aroma quality of strong-aroma baijiu.

Moreover, through the comparison of trace components between different brands of strong-aroma baijiu and the study on the change rule for trace components in the process of baijiu aging, it was found that there were similarities and differences between different brands of strong-aroma baijiu (Li et al., 2010; Song et al., 2020). On this basis, further exploration of the common and characteristic aroma compounds will provide a solid foundation for revealing the flavor code of strong-aroma baijiu, further refining national standards, promoting industry transformation and upgrading, and establishing grading and systematic geographical indications.

In conclusion, since 1964, the research on trace components in strong-aroma baijiu has been developing steadily. The research focus has gradually shifted from the preliminary exploration of the composition of trace compounds to the determination of the key aroma compounds by means of molecular sensory science. By far, a total of 141 main aroma compounds in strong-aroma baijiu have been clearly identified, including 46 esters, 22 aromatic compounds, 13 alcohols, 11 acids, 11aldehydes and ketones, 12 furans, 7 pyrazines, 8 acetals, 6 sulfur and nitrogen compounds and 5 lactones. The details are shown in the attached table (Table 3).

According to the above results of molecular sensory science, there are 32 compounds are identified as key aroma compounds in strong-aroma baijiu, including esters, alcohols, aroma compounds, sulfur compounds, nitrogen compounds, etc. Their structural formula and aroma characteristics are shown in Fig. 8. Esters, such as ethyl hexanoate, ethyl octanoate, ethyl butanoate and ethyl pentanoate, mainly expressing fruit aroma, which are considered as the primary aroma compounds of strong-aroma baijiu. Sulfur compounds and nitrogen compounds generally have a roasted, nutty aroma, and aromatic compounds contribute floral, honey and phenolic aroma. All these

[g] Odor thresholds was taken from reference (Fan and Xu, 2011).
[h] "-" means it's not found in the literatures.
[i] Odor thresholds was taken from reference (Dong et al., 2019).
[j] Odor thresholds was taken from reference (Zhao et al., 2018).
[k] Odor thresholds was taken from reference (Xu, 2019).

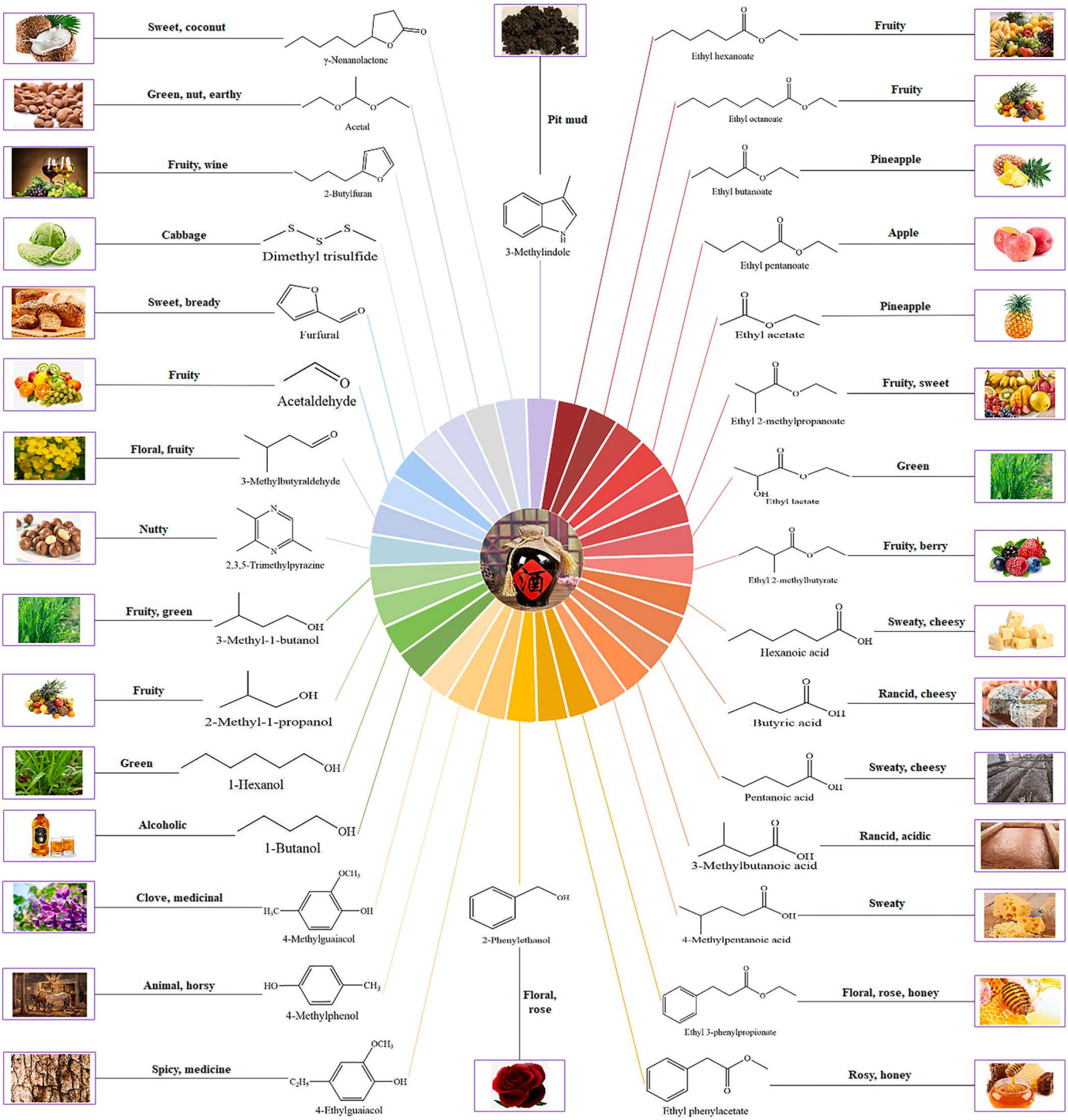

Fig. 8. Key aroma compounds in strong-aroma baijiu. According to the results of molecular sensory science, 32 kinds of aroma compounds are identified as key aroma compounds in strong-aroma baijiu. This figure shows the corresponding structural formulas and aroma characteristics of these 32 key aroma compounds.

compounds make strong-aroma baijiu a consistent and rich flavor.

6. Conclusion and prospect

Strong-aroma baijiu occupies a large part of the food industry in China. This paper systematically discusses the detection technology and the research progress of trace components in strong-aroma baijiu. With the development of scientific technology and in-depth studies, more and more trace components in strong-aroma baijiu have been found. At present, 141 trace components have been considered as the main aroma compounds of strong-aroma baijiu. Among them, 32 aroma compounds were identified as key aroma compounds, including 8 esters, 4 alcohols, 5 acids, 6 aromatic compounds, 1 sulfur compound, 1 nitrogen compound, 2 aldehydes, 2 furans, 1 pyrazine, 1 lactone, 1 acetal.

Up to now, the research on aroma profile of strong-aroma baijiu has made great progress, but there are still many problems in strong-aroma baijiu and even the whole baijiu industry. For example, the metabolic pathways and regulation of key aroma compounds in strong-aroma baijiu have not been thoroughly studied, and the interactions among some aroma compounds, as well as the olfactory and taste mechanisms of aroma compounds are still unclear.

Nowadays, the baijiu market is gradually becoming saturated, and

baijiu industry is confronted with an important transition and upgrading. While carry out scientific research on the baijiu is essential to steadily improve the quality of baijiu and promote the standardization, intelligence, and modernization of the baijiu industry. In the future, we should continue to improve and enrich the research means and tools of baijiu sensory omics, conduct in-depth research on the above-mentioned problems, construct the overall aroma profile of strong-aroma baijiu, and stabilize aroma quality of baijiu. This will lay the foundation for the improvement of related standards and the high-quality development of the baijiu industry.

CRediT authorship contribution statement

Junshan Wang: Investigation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Hao Chen**: Data curation, Writing – review & editing. **Yashuai Wu**: Data curation, Writing – review & editing. **Dongrui Zhao**: Writing – review & editing.

Conflict of interest statement

Any baijiu brand mentioned in this review does not involve commercial purposes. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China [32001826] the Science and Technology Program of Tibet Autonomous Region, China [XZ202001ZY0017N] and the project of the Science and Technology Department in Sichuan Province [2021ZYD0102].

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